

Skills Development for Adolescent Girls

Investment case, East Java, Indonesia

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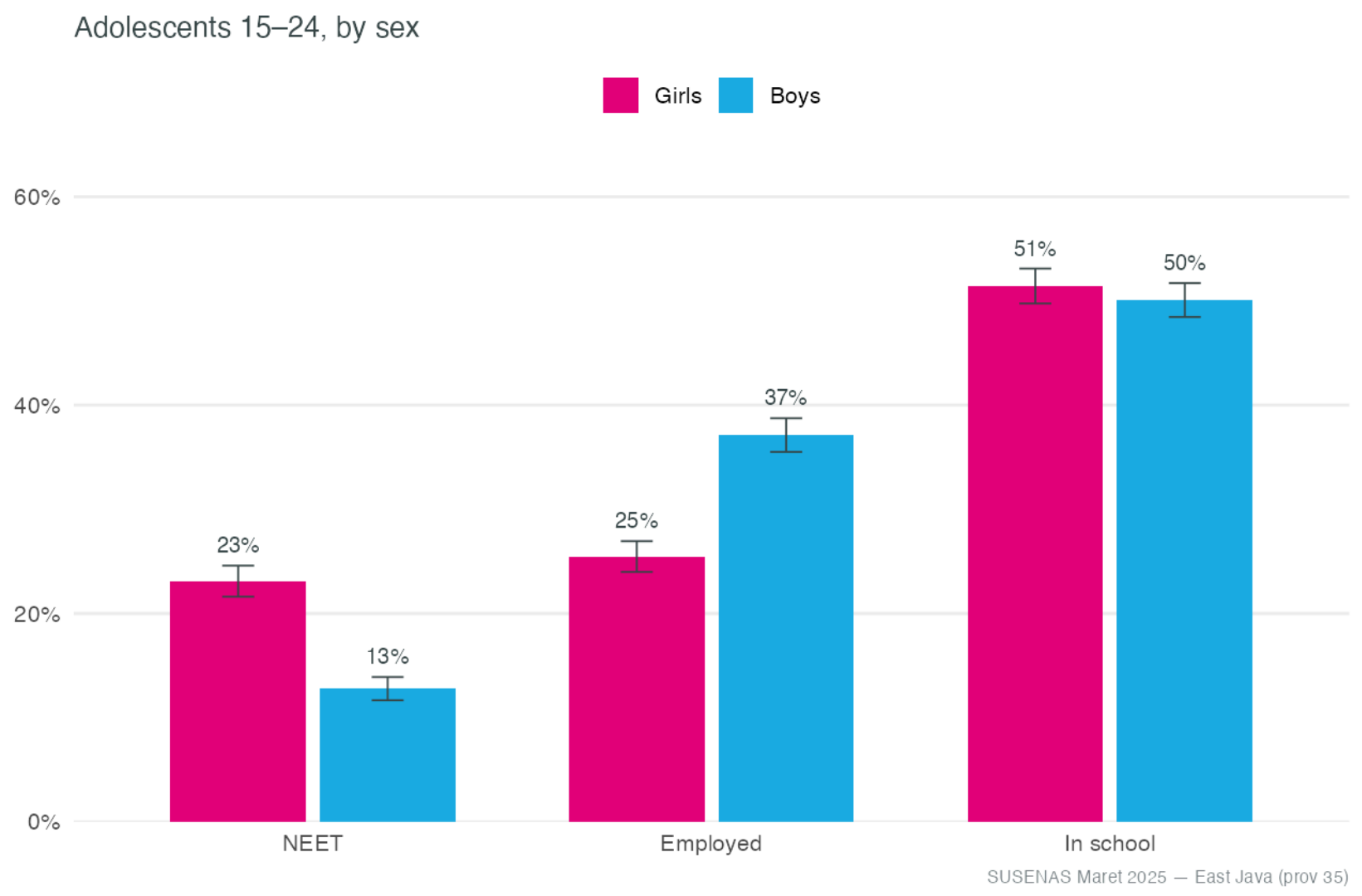
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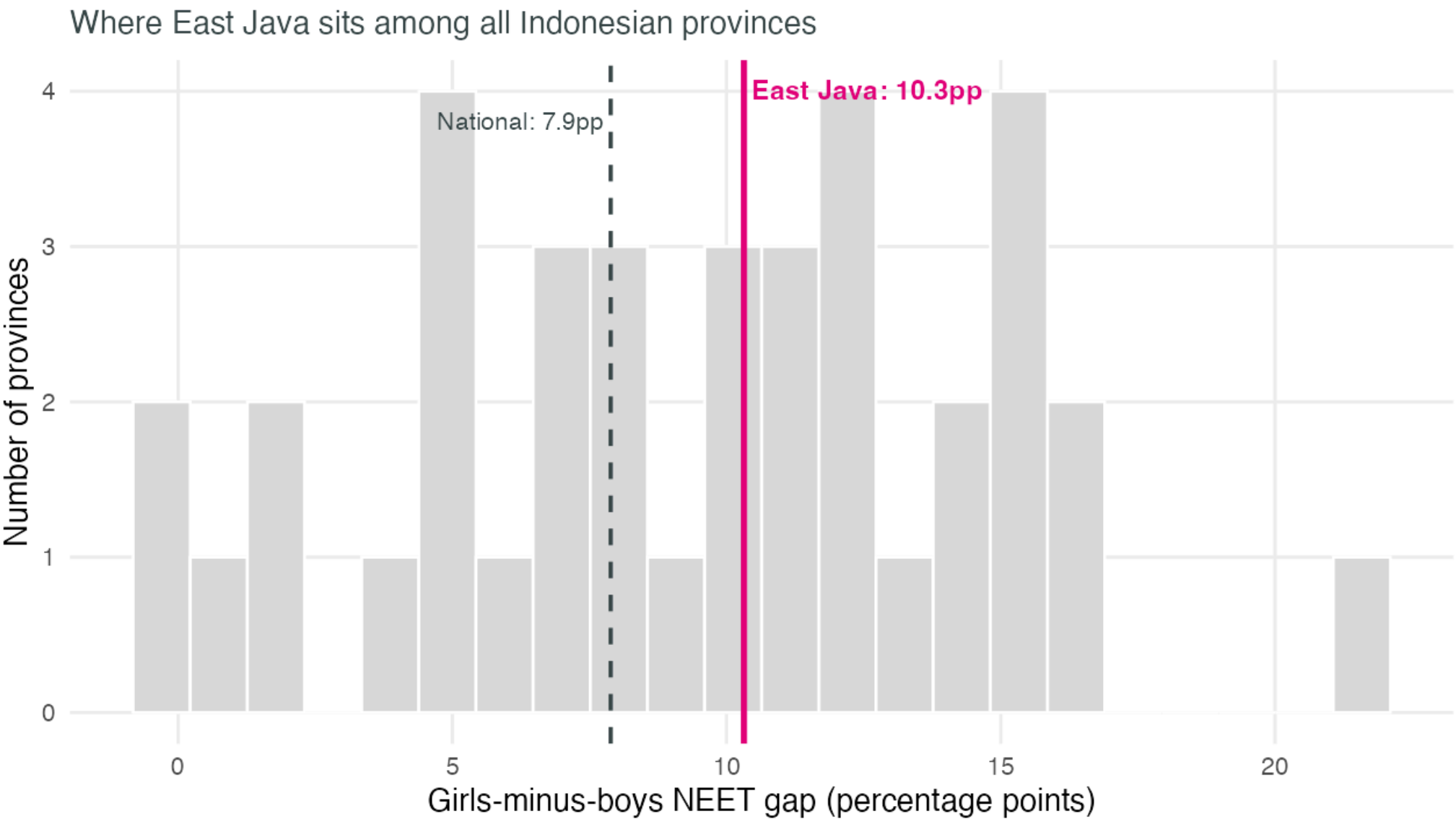
1 • The problem

Girls have the same school attendance rate as boys, but their NEET rate is close to double



NEET: not in employment, education or training. Girls 15–24: 51% in school (boys 50%), 23% NEET (boys 13%), 25% employed (boys 37%). The gap opens at the transition out of school. SUSENAS Maret 2025, survey-weighted, 95% CIs.

East Java's NEET gender gap is wider than the national average



SUSENAS Maret 2025, all provinces, survey-weighted; adolescents 15-24

Girls-minus-boys NEET gap: 10.3 points in East Java against 7.9 points nationally. Same survey, same definition, all provinces.

The gap costs an estimated USD 4.1 billion in forgone lifetime productivity

Forgone lifetime productivity from the NEET gender gap

Excess NEET girls (10.3pp gap above the boys' rate) x present value of lifetime earnings

Per excess-NEET girl (USD)

USD 14,143



Province-wide, today (USD millions)

USD 4,077.0M

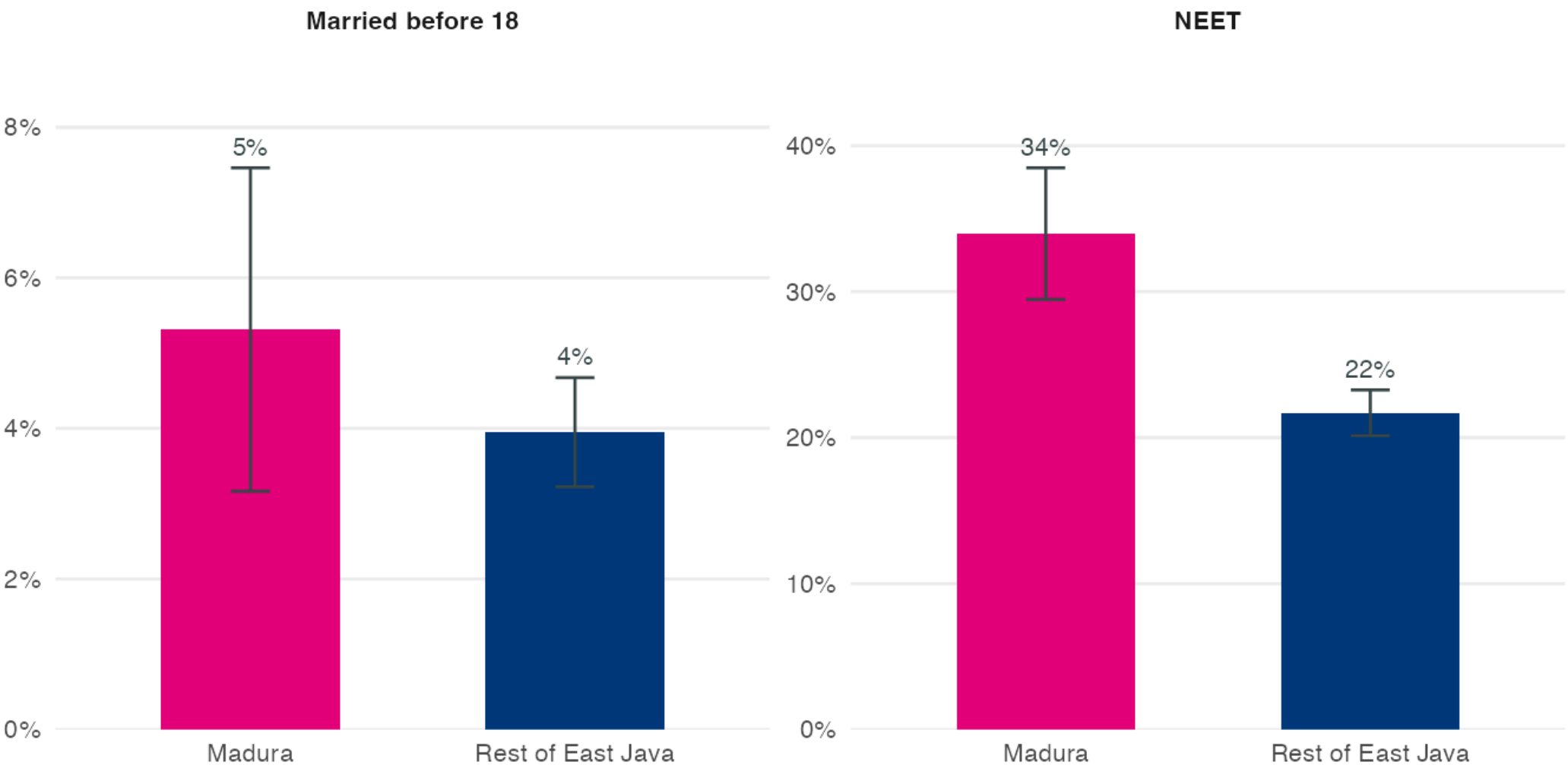


SUSENAS Maret 2025 — East Java (prov 35)

About 288,000 girls are NEET in excess of the boys' rate. Each would be expected to earn about USD 14,100 more over a working lifetime (present value) if she had the boys' NEET rate.

Madura has higher NEET for girls and boys, with a similar gender gap

Girls 15-24: Madura (Bangkalan, Sampang, Pamekasan, Sumenep) vs rest of East Java

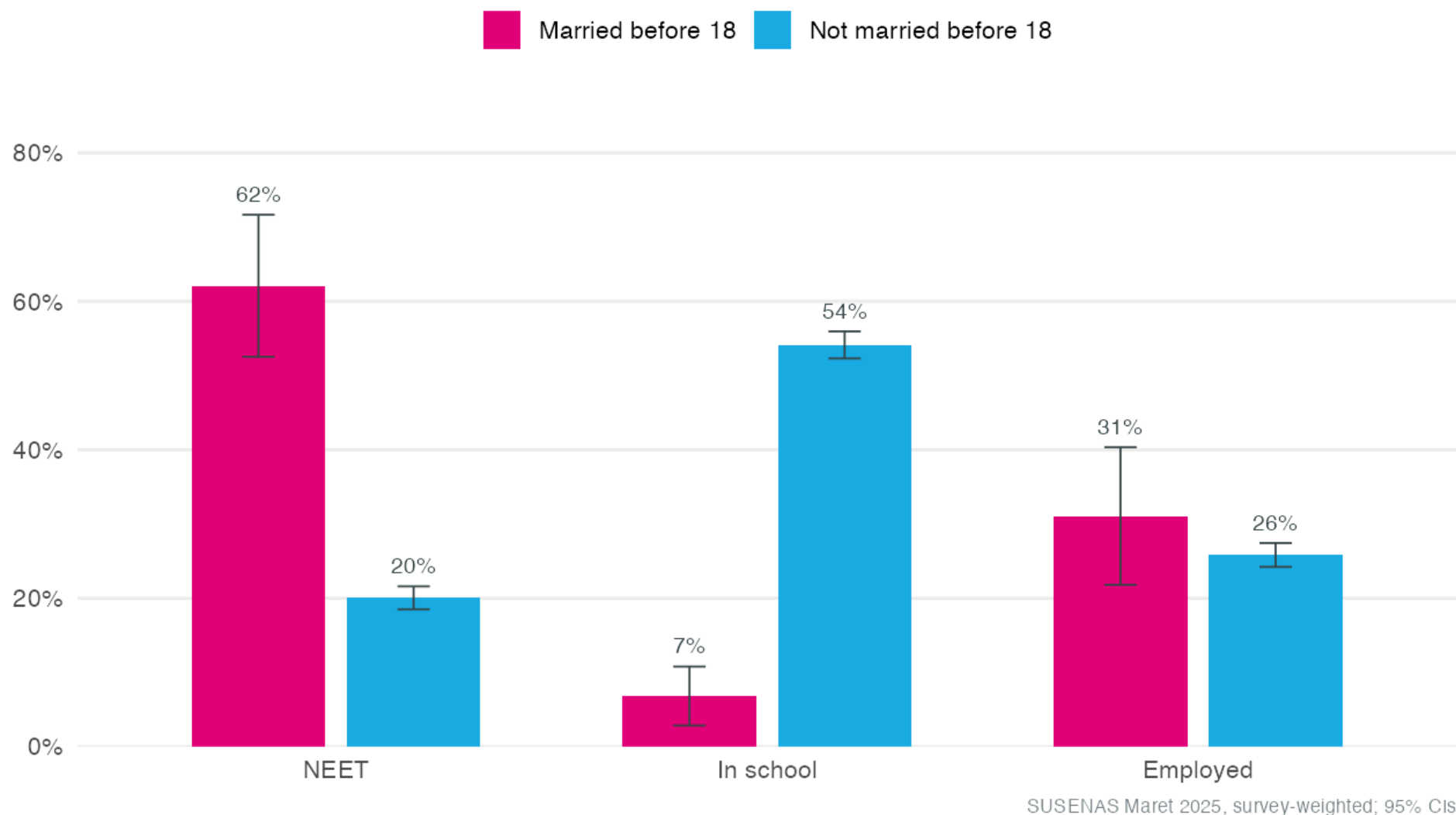


SUSENAS Maret 2025, survey-weighted; 95% CIs

Girls' NEET is 34% in Madura (Bangkalan, Sampang, Pamekasan, Sumenep) against 22% in the rest of East Java; boys' NEET is 23% against 12%. The gender gap is similar (11 vs 10 points), and fewer girls are in school (45% vs 52%). Madura is a priority area for a pilot.

Child marriage and outcomes: rest of East Java

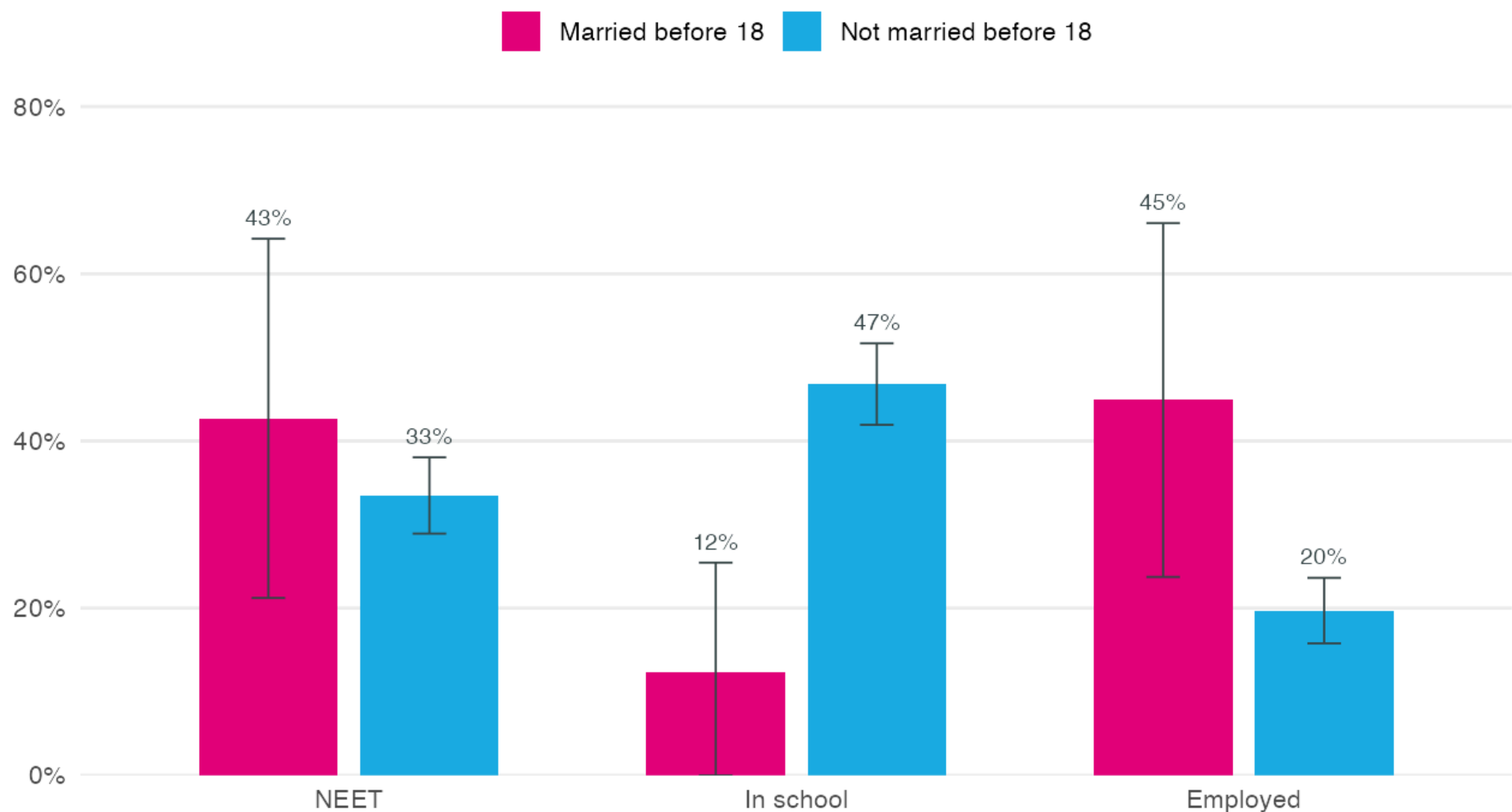
Girls 15-24, Rest of East Java



Girls 15–24 outside Madura, by whether they married before 18, survey-weighted (95% CIs). Regression tests by region are in the appendix. Early pregnancy before 18 overlaps almost entirely with child marriage in this data, so only child marriage is shown. Association, not causation.

Child marriage and outcomes: Madura

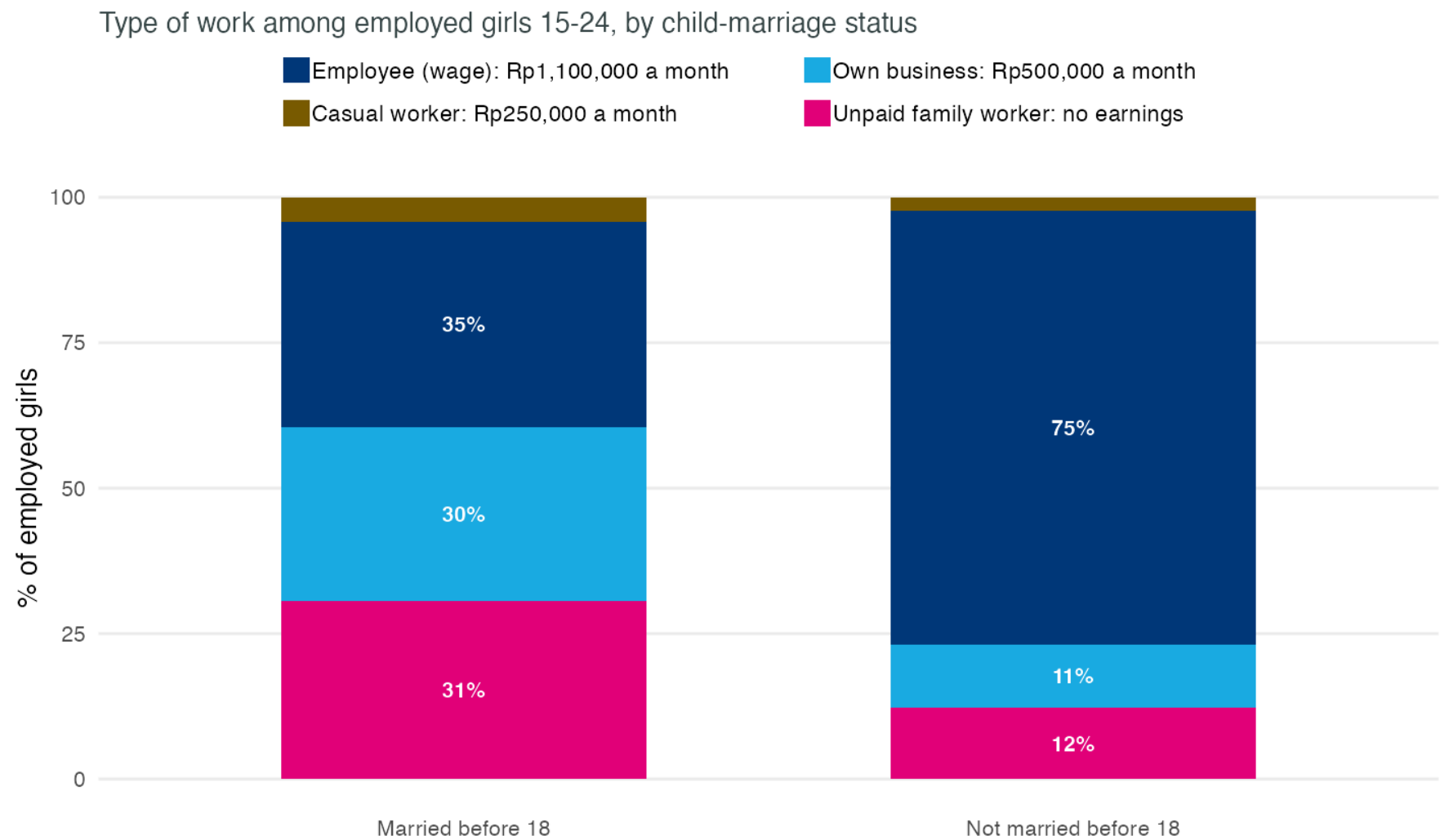
Girls 15-24, Madura



SUSENAS Maret 2025, survey-weighted; 95% CIs

Girls 15–24 in Bangkalan, Sampang, Pamekasan and Sumenep. Fewer girls, so wider intervals. Association, not causation.

Married girls who work do more unpaid family work and less wage employment

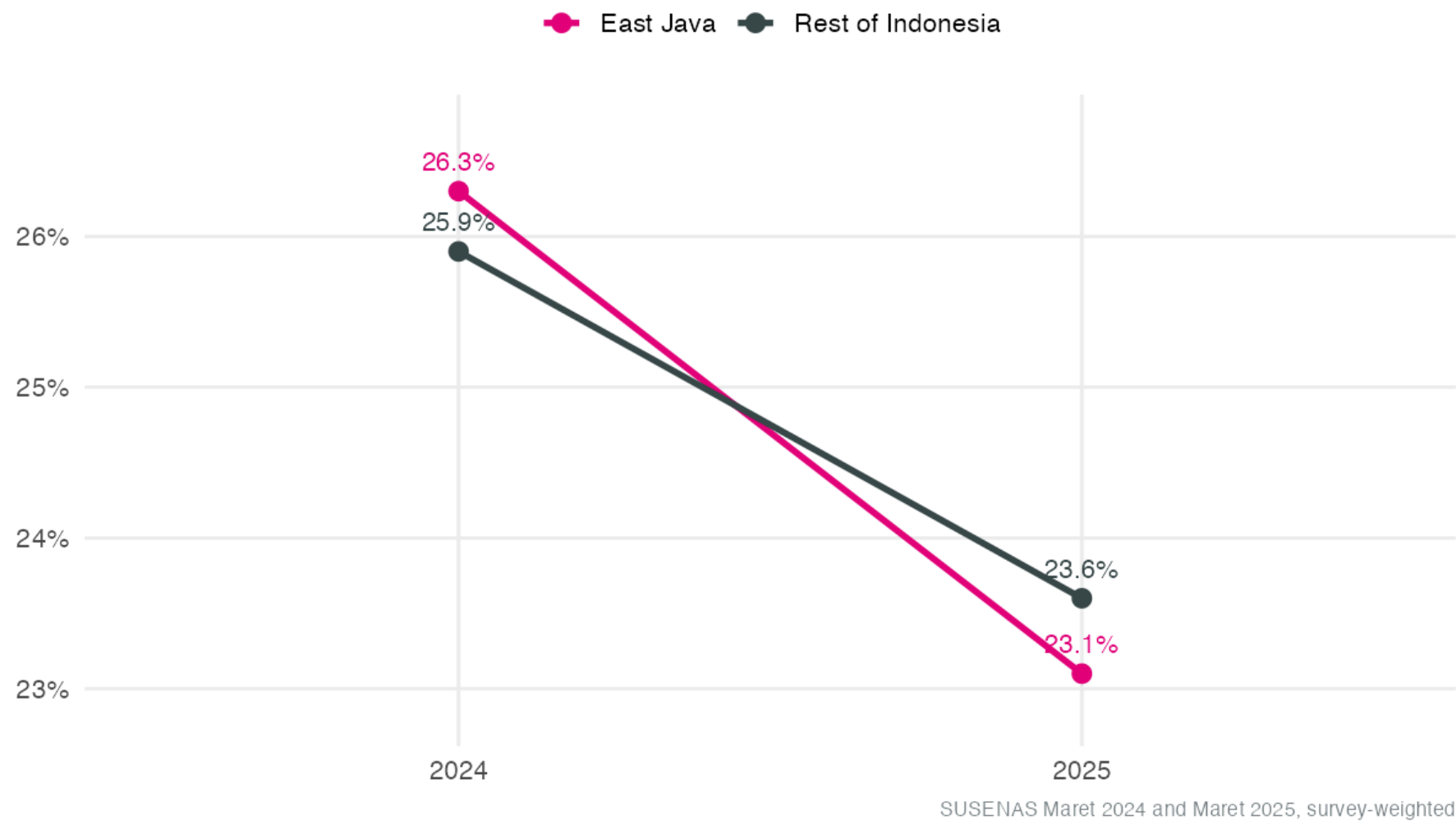


SUSENAS Maret 2025, East Java, survey-weighted. Legend: median monthly earnings of women 15-30 by type of work, IFLS5 (2014/15 prices).

Among employed girls married before 18, 35% are wage employees (others 75%) and 31% are unpaid family workers (others 12%); 22% work under 20 hours a week (others 12%). Based on 78 employed married girls, so indicative. SUSENAS has no earnings; the appendix shows earnings by type of work for young women from IFLS5.

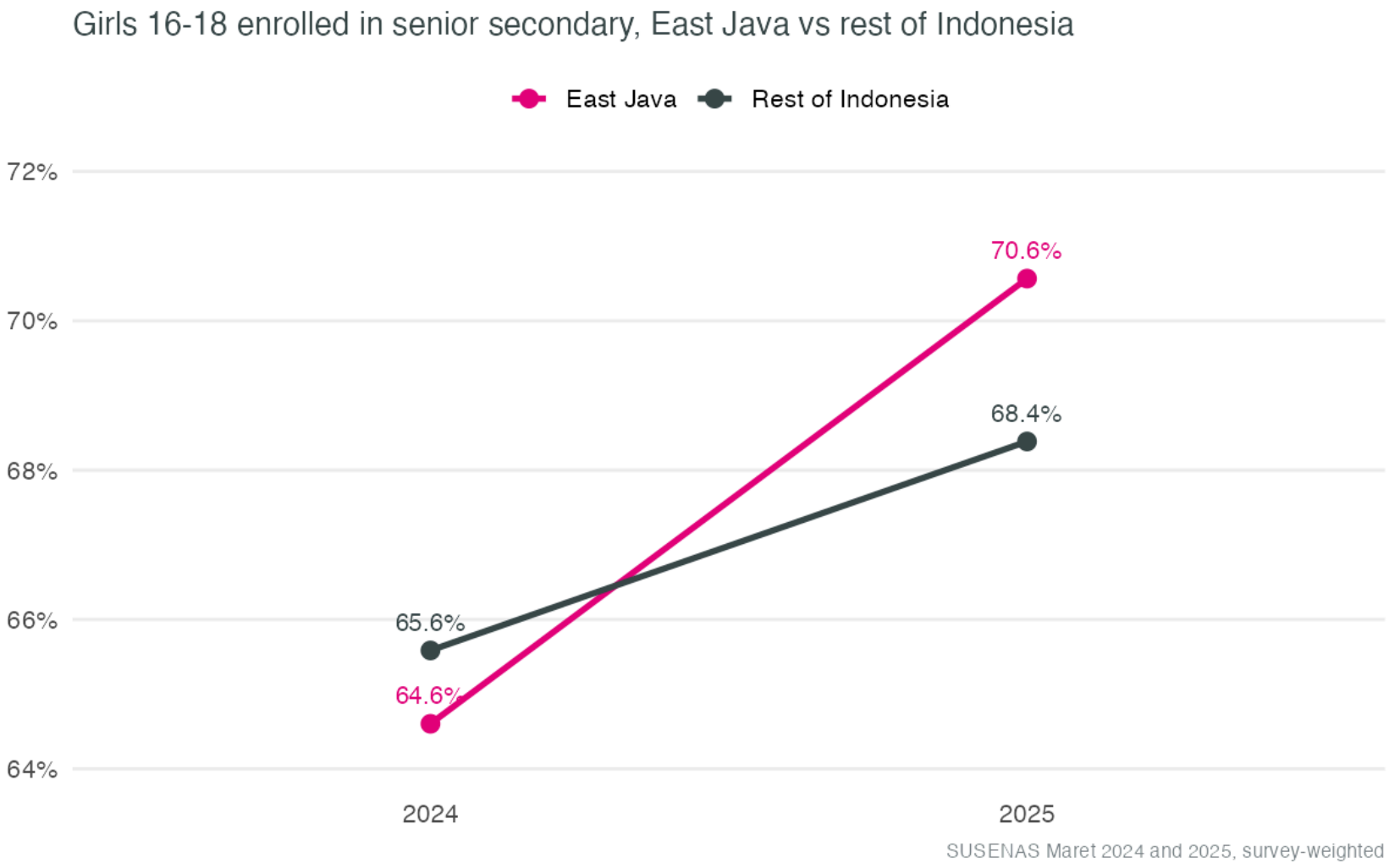
Girls' NEET in East Java and the rest of Indonesia, 2024 and 2025

Girls' NEET rate (15-24), East Java vs rest of Indonesia



Girls' NEET (15–24) fell 3.2 points in East Java and fell 2.3 points in the rest of Indonesia; child marriage fell 1.3 points and fell 0.7 points; tertiary enrolment at 18–20 rose 3.2 points and rose 5.0 points. Both years use the same definition (any work in the last week, or temporarily absent from a job; not attending school). Survey trends cannot separate a programme from wider changes; a comparison of programme and non-programme districts is pending.

Senior-secondary enrolment rose faster in East Java than in the rest of Indonesia

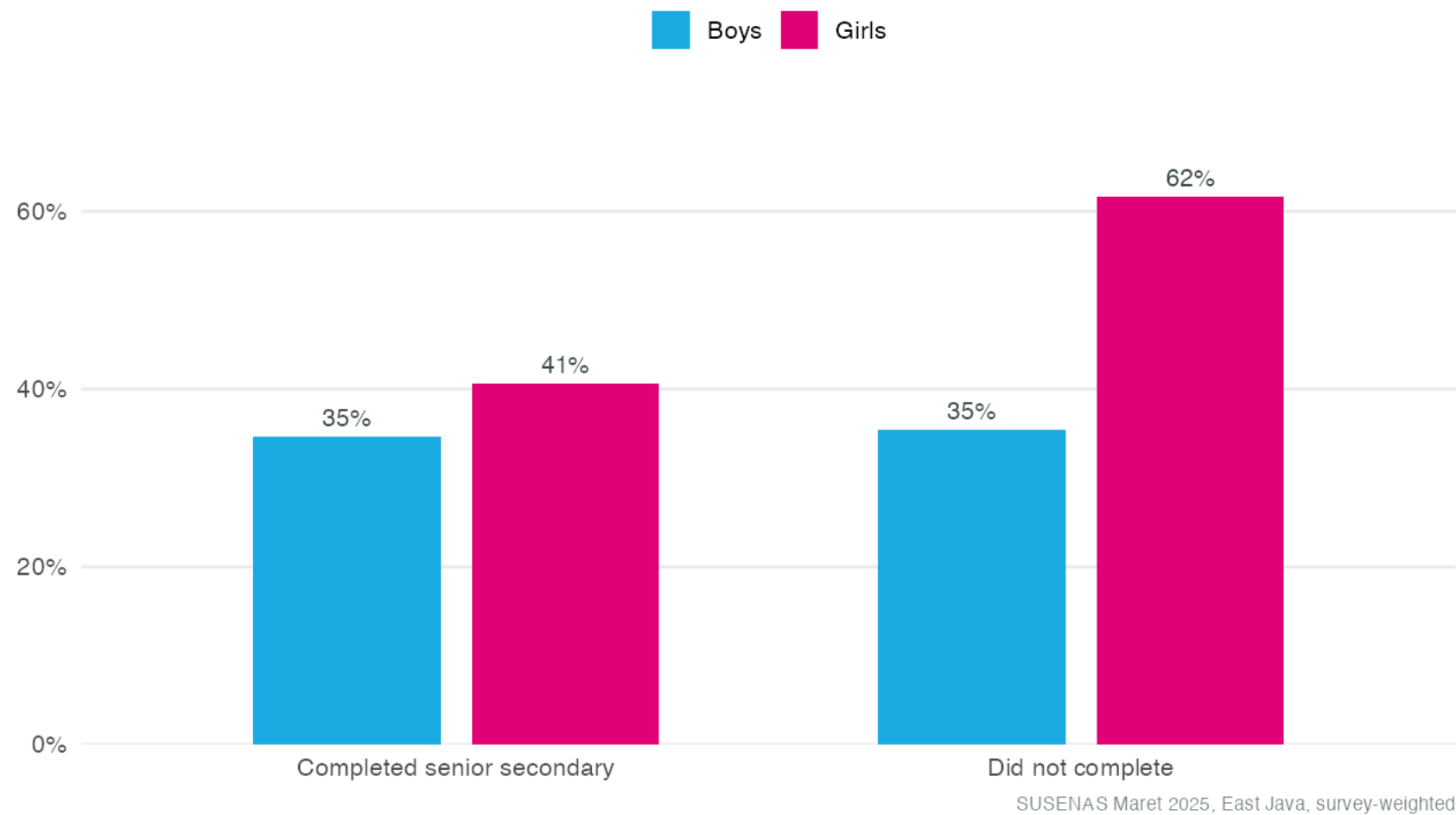


Share of 16–18 year olds enrolled in senior secondary rose 6.0 points for girls in East Java against 2.8 in the rest of Indonesia (boys: 5.8 against 2.6). The faster rise applies to both sexes, so it points to something affecting East Java’s schools generally rather than a girls-specific programme. The district comparison can test whether programme districts drove it.

2 · Theory of change

Completing senior secondary is associated with 20 points lower NEET for girls

NEET rate among 18-20 year olds who have left school, by senior-secondary completion



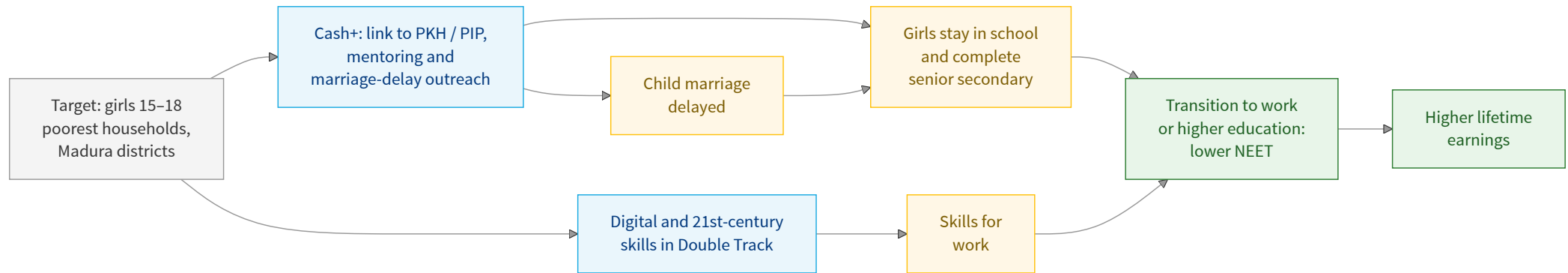
Among 18–20 year olds who have left school, girls who completed senior secondary have a NEET rate of 41%, against 62% for those who did not. Adjusted for age and area (girls 18–24), completion is associated with a 20-point lower NEET rate ($p < 0.001$). For boys there is no observed difference. Keeping girls in school to completion is the programme’s main lever.

Evidence from Indonesia

Programme	Design	Result	Relevance
PKH cash transfer (Priebe & Sumarto, 2025)	Quasi-experimental	Child marriage –3.5pp among girls	A cash component reduces child marriage
PIP education assistance (Ulfa & Rezki, 2024)	Matching and IV	School participation +11.4pp (pooled, not girls-specific)	Anchors the effect used in the ROI, with a transferability assumption
PIP, girls’ senior secondary (Caniago et al., 2021)	Observational regression	Girls’ senior-secondary enrolment +24.6pp (adjusted association)	Targets the stage where girls drop out
Vocational vs general secondary schooling (Pritadrajati, 2022)	Quasi-experimental	Female LFPR +3.4pp (significant at 10%); income +5.6% (not significant)	Used only as an uncertain scenario in the ROI

In SUSENAS 2025, girls in PKH households have lower child-marriage rates than other girls (3% vs 4–6%), even though PKH targets poorer households. Consistent with the PKH evidence, though not a causal estimate.

Theory of change



Each arrow rests on evidence from Indonesia: cash transfers reduce girls' child marriage (PKH); education assistance raises school participation (PIP, pooled) and girls' senior-secondary enrolment (PIP, adjusted association); completion is associated with 20 points lower NEET for girls (this analysis). The link from skills to earnings is the weakest: the Indonesian estimate is not statistically significant. Mentoring and outreach have only international evidence for girls. The cash component links girls to existing PKH and PIP rather than creating a new transfer.

3 · Return on investment

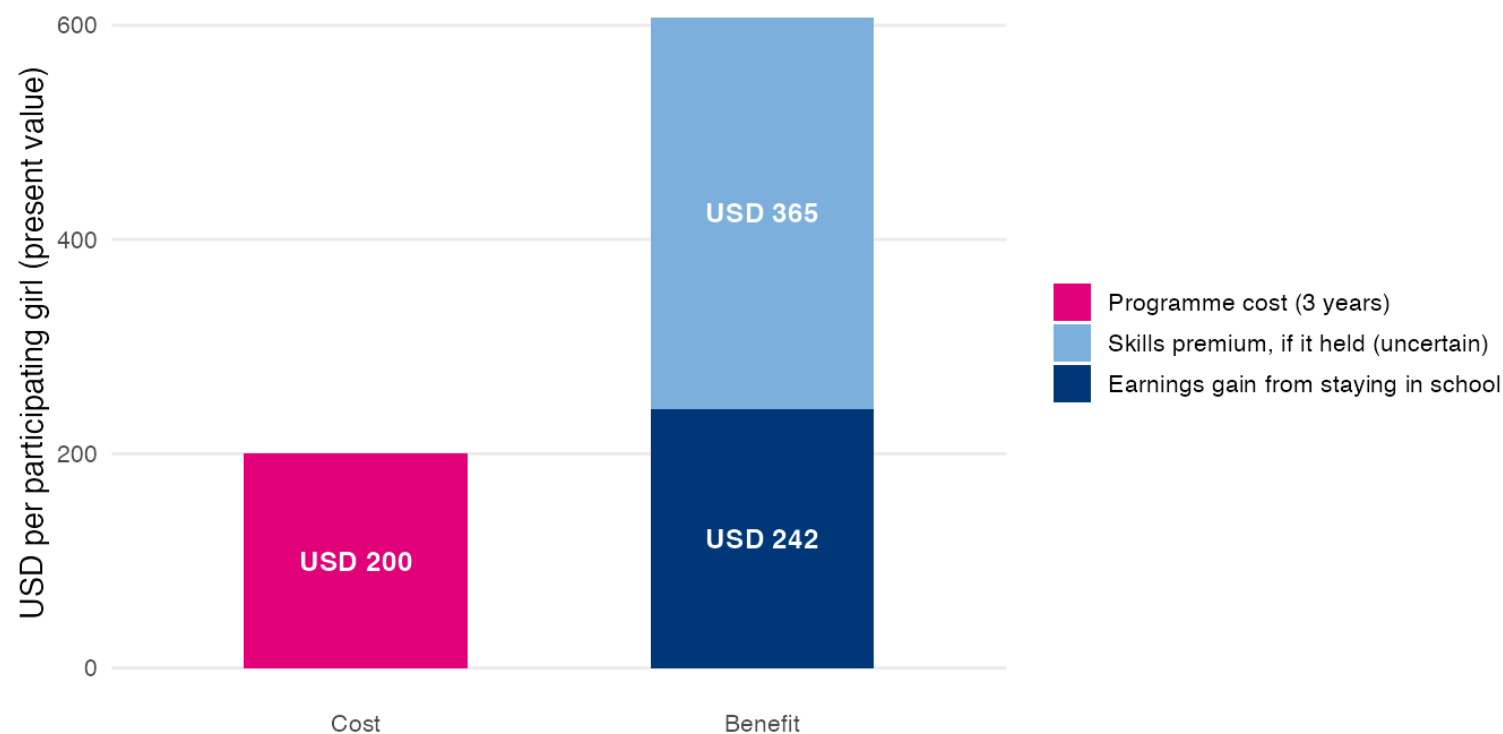
Key assumptions

Assumption	Value	Source
Reduction in girls' NEET	5 points	PIP attendance effect (+11.4pp), 45% assumed to come from girls who would be NEET
Extra schooling for a girl kept in school	3 years	Senior-secondary cycle
Earnings gain per year of schooling	8.0%	IFLS5 wage return for women, controlling for region, parents' education and selection into paid work
Female earnings	Rp20 million/year	Below East Java minimum wage 2025, reflecting informal work
Probability of earning (employment rate, women 25–54, East Java)	62%	SUSENAS Maret 2025
Earnings growth / discount rate	2% / 3% per year	Career growth; 3% real, with 2% and 5% in sensitivity
Skills earnings premium (scenario only, not in the central case)	5.6% for 10 years	Point estimate for vocational vs general schooling (Pritadrajati 2022), not statistically significant
Programme cost	USD 200 per girl	USD 58 per year for 3 years, plus 15% overhead; Cash+ links to existing PKH/PIP

How the return is built

Step	Value
Extra years of schooling for a girl kept in school	3
Earnings gain per year of schooling	8.0%
Earnings gain for that girl	24%
Female earnings (USD/yr) x labour force participation	1,266 x 62%
Extra earnings per year (USD)	188
Present value over the working life (USD per girl moved out of NEET)	4,843
Girls exposed per girl moved out of NEET (1 / NEET reduction)	20
Cost per girl moved out of NEET (USD)	4,002
Return per USD 1, retention channel (central)	1.21
Skills premium of 5.6% for all exposed girls who work, if it held (USD per exposed girl, PV)	365
Return per USD 1 if the skills premium held	3.04

Every USD 1 invested returns about USD 1.21 in lifetime earnings



Benefit-cost ratio by scenario

Scenario	Retention benefit (USD/girl)	Skills premium benefit (USD/girl)	Total benefit (USD/girl)	Cost (USD/girl)	BCR
Central: retention channel (Indonesian causal evidence)	242	0	242	200	1.21
Plus skills earnings premium of 5.6% (point estimate, not statistically significant)	242	365	607	200	3.04
Retention channel, earnings at the 2025 minimum wage	336	0	336	200	1.68
Retention channel, with Cash+ stipend for the poorest 40%	242	0	242	359	0.67

The central case is the retention channel; the second row adds the uncertain skills premium; the others show earnings at the 2025 minimum wage, and the cost of adding a Cash+ stipend for the poorest 40%. With other assumptions varied one at a time, the central return stays between USD 0.82 and USD 1.86 (appendix).

4 · Key messages

Key messages

1. Girls in East Java have the same school attendance rate as boys but are close to twice as likely to be NEET, a wider gap than nationally, costing about USD 4.1 billion in forgone lifetime productivity.
2. Completing senior secondary is associated with 20 points lower NEET for girls; child marriage with leaving school and more unpaid family work. Both are concentrated among poorer girls and in Madura.
3. A USD 200-per-girl package returns about USD 1.21 per USD 1 in lifetime earnings from keeping girls in school, before any benefit from skills, delayed marriage or health.

Policy recommendations

- 1. Target senior-secondary completion, not enrolment.** Girls already enrol at the same rate as boys; the loss happens at the transition. Set programme targets on completion and the first job, measured through SUSENAS.
- 2. Start in Madura and the poorest 40% of households.** NEET is highest and attendance lowest there, and child marriage falls steeply with household wealth, so each dollar does most there.
- 3. Link girls to PKH and PIP rather than creating a new transfer.** Indonesian evidence shows PKH reduces girls' child marriage; linkage uses existing systems. Cost a new stipend only if coverage beyond current entitlements is wanted.
- 4. Fund pedagogy and mentoring, not devices.** Internet use is already near universal, and devices alone did not raise learning in Uruguay.
- 5. Measure the programme's own effect.** Survey trends cannot separate the programme from wider changes. Compare programme and non-programme districts, and track whether girls move into wage employment rather than unpaid family work.

Appendix

Regression results: child marriage and outcomes

Hypothesis	Effect (pp)	SE	p	N
Child marriage → NEET	28.7	4.6	0.0000	6060
Child marriage → in school	-27.5	3.0	0.0000	6060
Child marriage → employment	-1.2	4.6	0.7942	6060

Linear-probability models, girls 15–24, survey-weighted, PSU-clustered. **Controls: age and urban/rural.** Poverty is not controlled for because the 2025 expenditure module is not yet available. The association with employment is small and not statistically significant. Association, not causation.

Child marriage and outcomes by region: regression results

Region	Outcome	Effect (pp)	SE	p	N
Madura	Employed	18.7	11.1	0.093	664
Madura	NEET	3.1	12.3	0.801	664
Madura	In school	-21.8	9.8	0.026	664
Rest of East Java	Employed	-4.6	5.1	0.361	5396
Rest of East Java	NEET	33.2	5.0	0.000	5396
Rest of East Java	In school	-28.5	3.1	0.000	5396
All East Java	Employed	-1.2	4.6	0.794	6060
All East Java	NEET	28.7	4.6	0.000	6060
All East Java	In school	-27.5	3.0	0.000	6060

Linear-probability models, girls 15–24, survey-weighted, PSU-clustered; controls for age and urban/rural. Effects in percentage points.

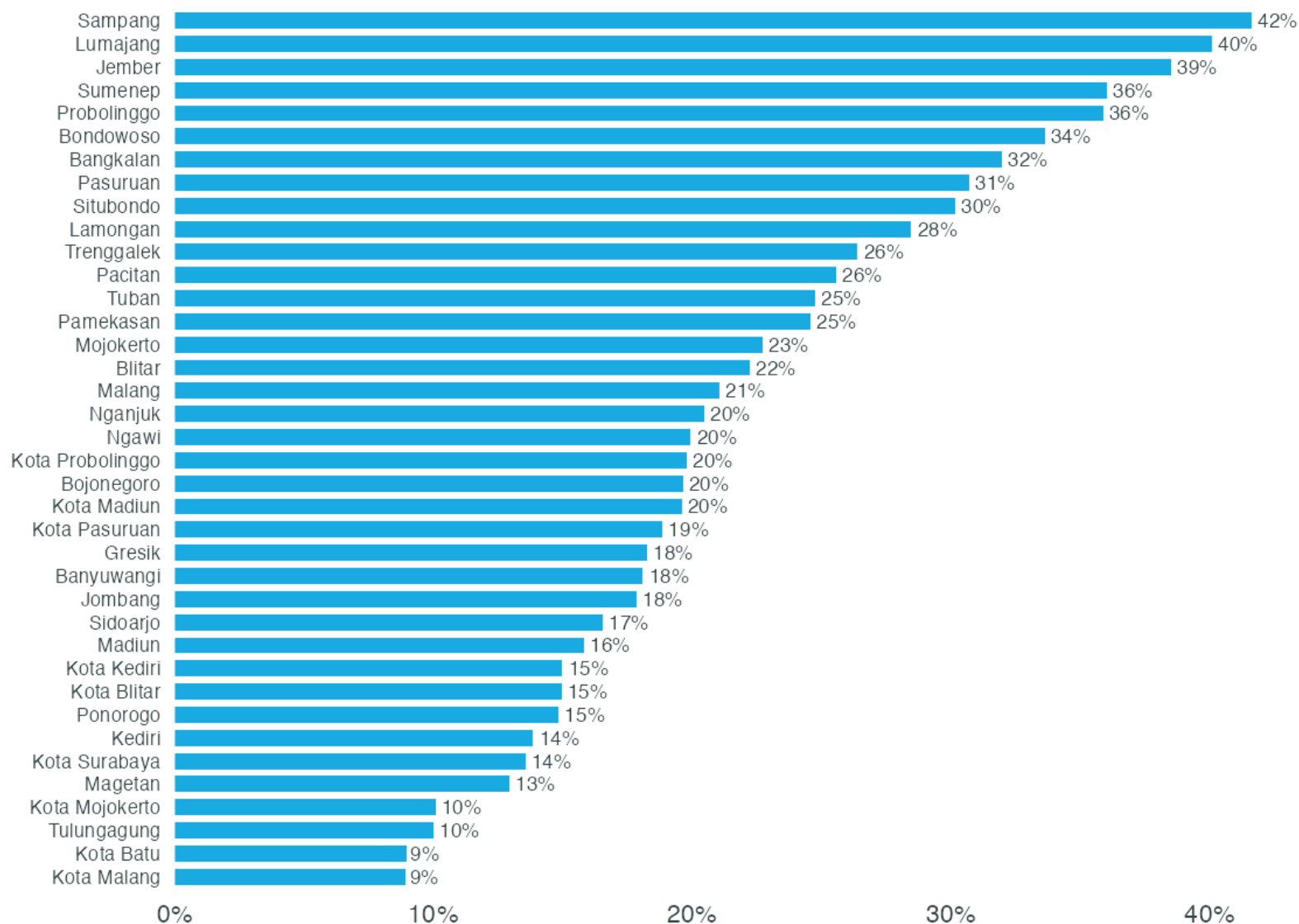
Earnings by type of work, young women (IFLS5)

Type	Women 15-30	With earnings reported	Median monthly earnings (IDR)
Employee (wage)	1662	1652	1,100,000
Own business	689	667	500,000
Casual worker	203	203	250,000
Unpaid family worker	614	614	0

Women 15–30, IFLS5 (2014/15 prices). SUSENAS collects no earnings. Wages for employees and casual workers; net profit for own businesses; unpaid family workers have no earnings by definition.

NEET among girls, by district

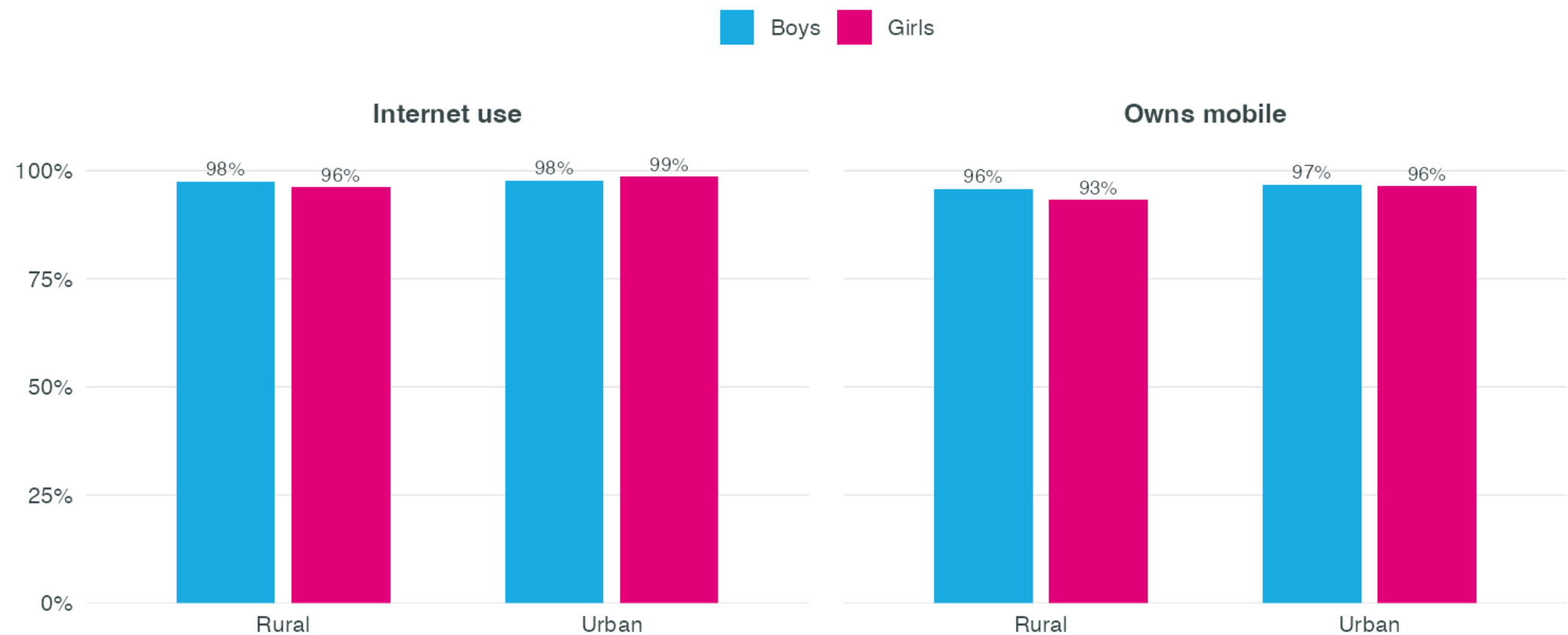
Girls' NEET rate, by East Java district (≥15 sampled girls)



Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Digital access and use

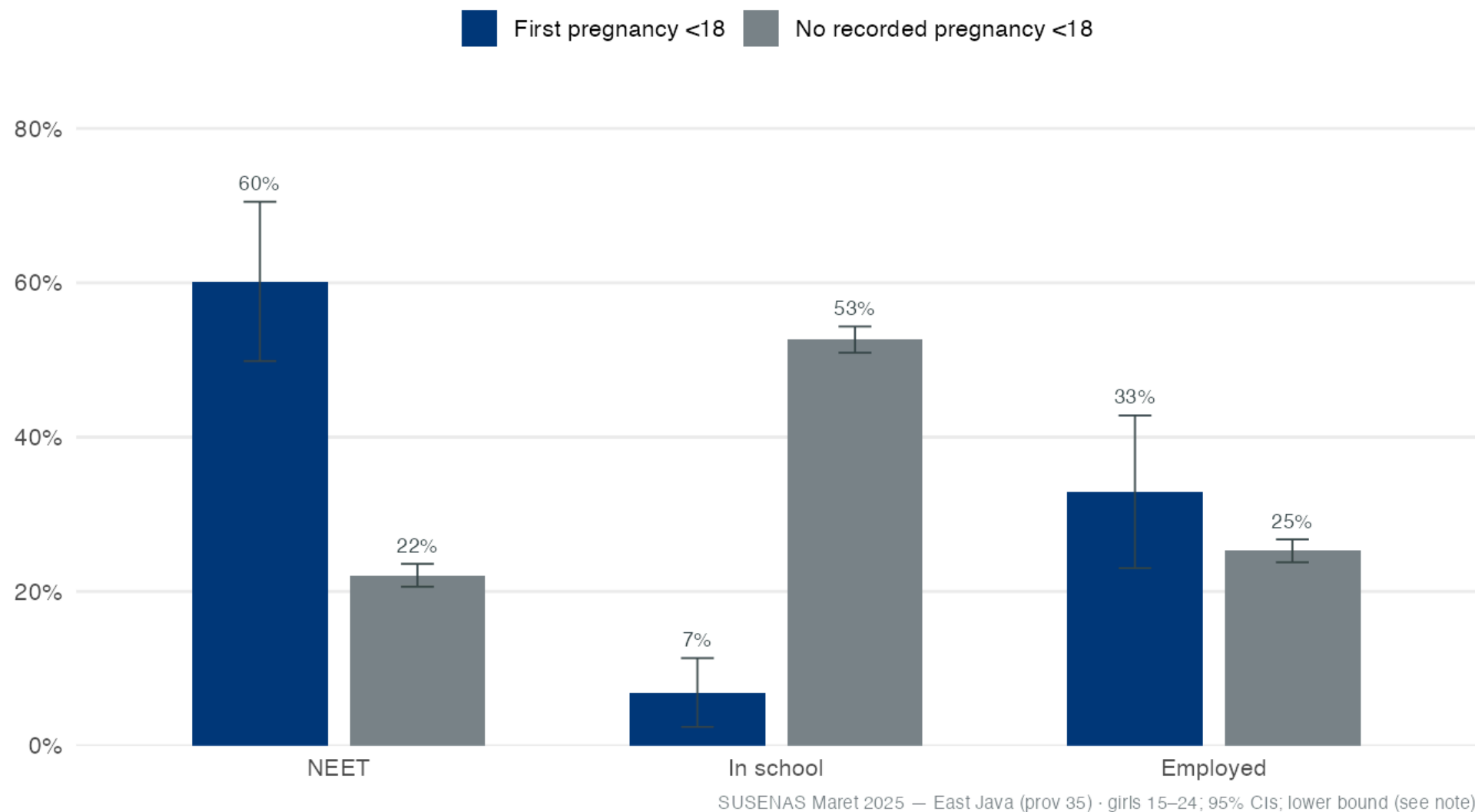
Digital access and use, by sex and area



Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Early pregnancy and outcomes

Girls with an early pregnancy are far likelier to be NEET and out of school



Recorded only for ever-married women, so a lower bound. Girls with a first pregnancy before 18 are almost all girls married before 18, so the main deck reports child marriage only.

Child marriage and NEET by household PKH receipt

region	PKH	n	Married_u18	NEET
Madura	Not PKH	509	6.1	35.9
Madura	PKH household	155	3.0	28.5
Rest of East Java	Not PKH	4661	4.1	22.1
Rest of East Java	PKH household	735	3.2	19.5

Girls 15–24, SUSENAS 2025. PKH targets poorer households, so lower rates among recipients are consistent with a protective effect, but are not a causal estimate.

Cost components

Component	USD/girl/yr	Source
Digital content & platform	10	Kemendikbudristek Merdeka Mengajar platform; INOVASI digital-learning unit costs
Teacher training (annualised)	15	Double Track PD days, annualised
Mentoring & socio-emotional	10	ELA/Skills4Girls facilitator costs
School connectivity/devices (shared)	15	Shared labs, no 1:1 devices
Ancillary: marriage-delay + flexible learning	8	ToR guidance: outreach to delay marriage, flexible options for caregivers
Annual direct cost per girl	58	Sum
Total per girl (3 years + 15% overhead)	200	Programme cost per girl

These are benchmark unit costs from comparable programmes, not quotes from Double Track; they are placeholders until the provincial government shares actual budgets. The transfer itself is not costed because the Cash+ component links girls to existing PKH and PIP entitlements; a new stipend for the poorest 40% at the PIP senior-secondary rate would add about USD 46 per girl per year (ROI table).

Sensitivity of the return to each assumption

Parameter	BCR at low value	BCR at high value
Skills earnings premium (all exposed girls who work)	1.21	3.04
NEET reduction per exposed girl	0.61	1.82
Mincer return per year of edu	0.85	1.86
Discount rate	0.82	1.49
Annual cost per girl (sensitivity)	1.56	0.94
Female annual wage (East Java)	0.91	1.51
Real earnings growth	0.90	1.42
Female labour force participation	0.87	1.26
Years the skills premium lasts	1.21	1.21

Each assumption varied on its own, others held at central values.

- **Low values:** a 2.5-point NEET effect (half the PIP-based estimate), the consumption-based return of 5.6% instead of 8%, earnings of Rp15m instead of Rp20m, a 5% discount rate, or a cost of USD 75 a year instead of 58.
- **High values:** a 7.5-point NEET effect, the 12.2% return without the selection correction, earnings of Rp25m, a 2% discount rate, or a cost of USD 45 a year.
- The effect size and the return to schooling matter most; no single assumption on its own takes the central case far from break-even.

What the return counts

Counted

Higher schooling leads to a lifetime earnings premium for girls moved out of NEET, using the female return to schooling over a discounted working life.

Excluded (so the return is a lower bound)

Delayed child marriage · health and productivity spillovers · intergenerational gains · fiscal returns · broader empowerment

Comparable international evidence

Study	Design	Sample	Finding	Relevance for East Java
Bergstrom & Özler 2021	Review of 104 interventions from 70 studies	Adolescent girls	Mixed evidence for life-skills training and girls' clubs; stronger for cash and schooling support	Supports cash and schooling components; cautions on clubs alone
Bandiera et al. 2018 (Sierra Leone)	RCT	Girls 12–25	Enrolment +8.5pp during the Ebola period; effects on income-generating activity vary by age	Girls' clubs protected schooling in a crisis context
Aramburu, Goicoechea & Mobarak 2021 (Argentina, Colombia)	RCT	Women 18–50	Coding bootcamps raised coding skills	Digital-skills content
Näslund-Hadley et al. 2025 (Paraguay)	RCT	Second-grade pupils	+0.09 SD computational thinking	Pedagogy; limited transfer to adolescent employment
Gülen et al. 2025 (Turkey)	Nonrandomized controlled pre/post	Girls 14–16	Higher STEM aspirations and motivation	Girls' STEM engagement
Prakash et al. 2019, Samata (India)	RCT	Poor girls 13–16	No overall effect on completion or marriage; schooling gains in one of two districts	Transition support alone had limited effects
Edmonds et al. 2021, Room to Read (India)	RCT	Girls	Dropout 4.1pp lower (main survey estimate)	Mentoring and life skills reduce dropout
Ashraf et al. 2025, Ten Years of Relational Power (Zambia)	RCT, 10-year follow-up	Girls ~14	+0.23 years of schooling	Durable, low-cost soft-skills layer
de Melo et al. 2014, Plan Ceibal (Uruguay)	Quasi-experimental	Pupils	No detectable maths or reading gains in the first two years	Devices alone did not raise learning; fund pedagogy
Faizah, Isbah & Azca 2025 (online; 2026 issue)	Qualitative	Indonesian youth	Norms, care duties and marriage drive girls' NEET	Indonesia-specific NEET context

Findings as reported in each source, with design and sample as verified. Used to inform programme design; the ROI effect comes from the Indonesian evidence. References on the last slide.

Returns to schooling: specification and results

Wage-based (IFLS5 2014/15, adults 20–60, estimated separately for women and men, weighted):

$$\ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 \text{Exp}_i + \beta_3 \text{Exp}_i^2 + \beta_4 \text{Urban}_i + \gamma_p + \phi_f + \phi_m + \delta \lambda_i + \varepsilon_i$$

w : monthly wage; S : years of schooling; Exp: potential experience (age minus schooling minus 6); γ_p : province fixed effects; ϕ_f, ϕ_m : father’s and mother’s education (fixed effects); λ : selection term from a first-stage model of working, in which children under 5 in the household predict whether a woman works but not her wage (Heckman two-step).

Sex	Model	Return per year	SE	N
Women	(1) Baseline	12.0%	0.5	3830
Women	(2) + Region	12.2%	1.1	3830
Women	(3) + Family background	10.9%	1.1	3828
Women	(4) + Selection (Heckman)	8.0%	1.2	3828
Men	(1) Baseline	9.4%	0.3	6844
Men	(2) + Region	8.7%	0.6	6843
Men	(3) + Family background	8.0%	0.5	6842
Men	(4) + Selection (Heckman)	10.1%	1.0	6842

The ROI uses the women’s estimate with all controls (model 4, 8.0%). The consumption-based return from SUSENAS 2024 (5.6% for women; household per-capita expenditure regressed on schooling, schooling², age, age² and urban) is the low case in the sensitivity analysis.

Data sources

- **SUSENAS Maret 2025** (BPS): NEET, school attendance and completion, child marriage, early pregnancy, type of work and hours, PKH receipt and digital access, for East Java and all provinces. East Java: 98,738 individuals, 12,599 aged 15–24, of whom 6,060 girls.
- **SUSENAS Maret 2024** (BPS): the 2024 comparison, and per-capita expenditure for the consumption-based return to schooling.
- **IFLS5 2014/15**: wages for the wage-based return to schooling.
- **Sakernas** (BPS): female labour force participation in East Java.
- **UMP Jawa Timur 2025**: benchmark for female earnings.
- **Published evaluations** of PKH, PIP and vocational training: the programme effect used in the return on investment.

All SUSENAS estimates are survey-weighted, with PSU/strata-clustered standard errors.

Definitions

- **NEET (15–24):** not in employment, education or training, built as not attending school (R611; 2024: R610) and not working in the last week or temporarily absent from a job (R703_A; R704, 2024: R705).
- **Married before 18:** age at first marriage (R409) under 18.
- **Senior-secondary completion:** highest certificate (R615) at SMA/SMK/MA level or above.
- **Type of work:** status in main job (R706); hours (R707).
- **Madura:** Bangkalan, Sampang, Pamekasan, Sumenep.

Several variables were renumbered between the 2024 and 2025 questionnaires (school attendance R610 to R611, temporarily absent R705 to R704, certificate R614 to R615, employment status R707 to R706); the activity block (R703) is the same in both years and both years are coded identically. The operational NEET definition does not separately exclude participants in non-formal training, which SUSENAS does not record.

References

- Aramburu, J., Goicoechea, A. and Mobarak, A.M. (2021). **Coding Bootcamps for Female Digital Employment: Evidence from an RCT in Argentina and Colombia**. World Bank.
- Ashraf, N., Bau, N., Low, C. and McGinn, K. (2020). Negotiating a Better Future: How Interpersonal Skills Facilitate Intergenerational Investment. **Quarterly Journal of Economics**, 135(2).
- Ashraf, N. et al. (2025). **Ten Years of Relational Power**. Long-run follow-up, American Economic Association.
- Bandiera, O., Buehren, N., Goldstein, M., Rasul, I. and Smurra, A. (2018). **The Economic Lives of Young Women in the Time of Ebola: Lessons from an Empowerment Program**. World Bank.
- Bergstrom, K. and Özler, B. (2021). **Improving the Well-Being of Adolescent Girls in Developing Countries**. World Bank Policy Research Working Paper.
- Caniago, et al. (2021). Programa Indonesia Pintar and girls' senior-secondary enrolment. Universitas Negeri Semarang journal, Table 5.
- de Melo, G., Machado, A. and Miranda, A. (2014). **The Impact of a One Laptop per Child Program on Learning: Evidence from Uruguay**. IZA Discussion Paper.
- Edmonds, E., Feigenberg, B. and Leight, J. (2021). **Advancing the Agency of Adolescent Girls**. Working paper; main survey estimate.
- Faizah, Isbah, M.F. and Azca, M.N. (2025, online; 2026 issue). Gendered transitions and NEET youth in Indonesia. SAGE.
- Gülen, S. et al. (2025). Robotics and coding for adolescent girls, Turkey. SAGE.
- Näslund-Hadley, E. et al. (2025). **Irumi: computational thinking in early primary, Paraguay**. Inter-American Development Bank.
- Prakash, R. et al. (2019). Samata: secondary-school transition for adolescent girls, India. **Journal of Global Health Reports**.
- Priebe, J. and Sumarto, S. (2025). Program Keluarga Harapan and child marriage in Indonesia. Elsevier.
- Pritadrajati, D. (2022). Vocational and general secondary schooling and labour-market outcomes in Indonesia. **Bulletin of Monetary Economics and Banking**.
- Ulfa, R. and Rezki, J.F. (2024). Programa Indonesia Pintar and school participation. Universitas Negeri Semarang journal, Table 4.
- Data.** BPS, Survei Sosial Ekonomi Nasional (SUSENAS) Maret 2024 and Maret 2025. Strauss, J., Witoelar, F. and Sikoki, B. (2016). **The Fifth Wave of the Indonesia Family Life Survey (IFLS5)**. RAND. Upah Minimum Provinsi Jawa Timur 2025.